

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

You can support your child to learn more about the interior angles of triangles, quadrilaterals and regular polygons with this activity. There's an information sheet which you can use to help your child to learn about the properties of these shapes, followed by a set of unknown angles activities for each one. No protractor necessary!

What skills does this practise?

Properties of 2D Shapes

Interior Angles

Further Activity Ideas and Suggestions

For more maths learning, you can use these [Number and Place Value Challenge Cards](#) to test your child's understanding of place value, learn more about negative numbers with this [overview and maze pack](#) or practise addition and subtraction with this [activity booklet](#).

Parents Blog



Twinkl Kids' TV



Homework Help



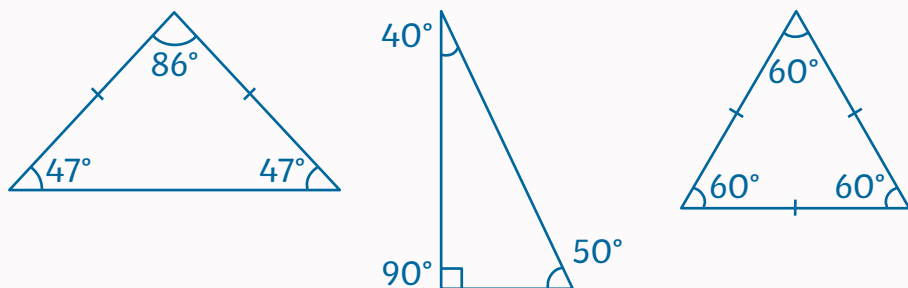
twinkl

Parents Hub

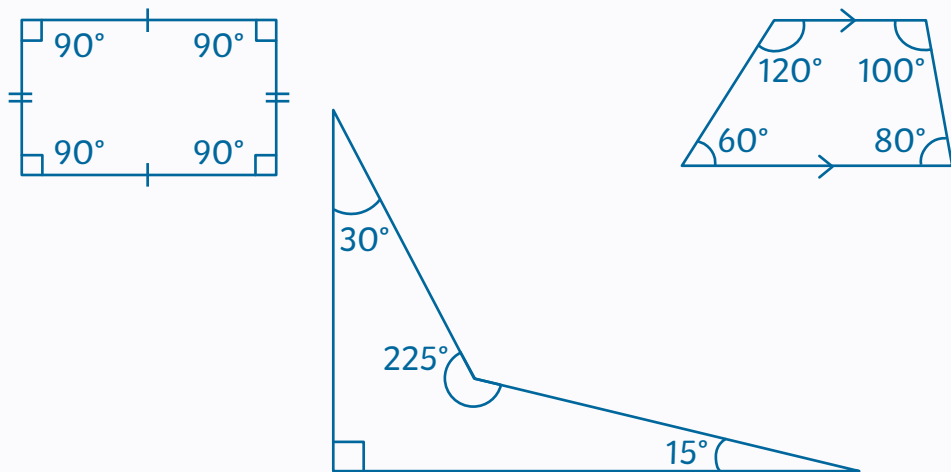
Angle Facts

Once you have learned some basic facts about angles, you will be able to solve a variety of problems. Here are some useful facts.

The interior angles of a **triangle** always add up to 180° .

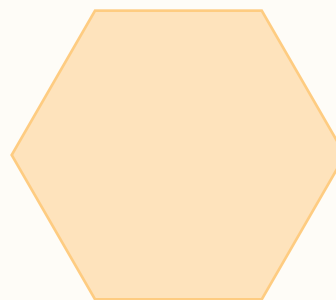


The interior angles of a **quadrilateral** always add up to 360° .



The sum of the interior angles of a **regular polygon** can be found by solving $(n - 2) \times 180^\circ$, where n = number of sides.

We can then divide the sum of the interior angles by the number of sides to find what each individual angle measures.



A hexagon has 6 sides.

$$n = 6$$

$$(6 - 2) \times 180^\circ = 4 \times 180^\circ = 720^\circ$$

The sum of interior angles = 720° .

$$720^\circ \div 6 = 120^\circ$$

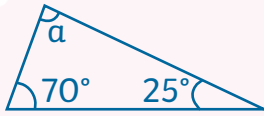
Each individual angle = 120° .

Angles in Triangles

Use your knowledge of angles to find the missing angles without using a protractor.

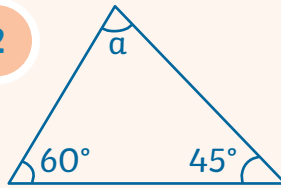
The interior angles of a triangle always add up to 180° .

1



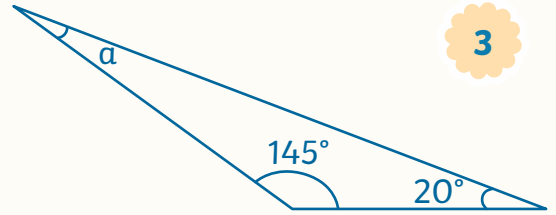
$$a = \underline{\hspace{2cm}}$$

2



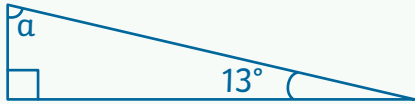
$$a = \underline{\hspace{2cm}}$$

3



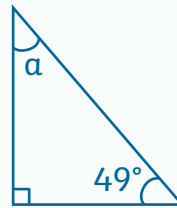
$$a = \underline{\hspace{2cm}}$$

4



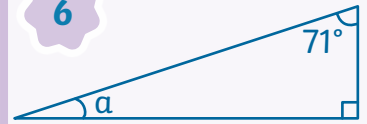
$$a = \underline{\hspace{2cm}}$$

5



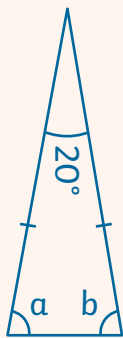
$$a = \underline{\hspace{2cm}}$$

6



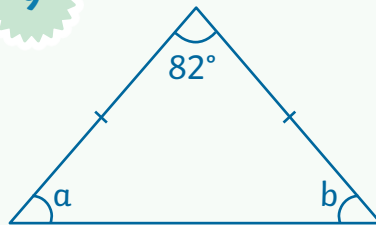
$$a = \underline{\hspace{2cm}}$$

8



$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

9



$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

10



$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

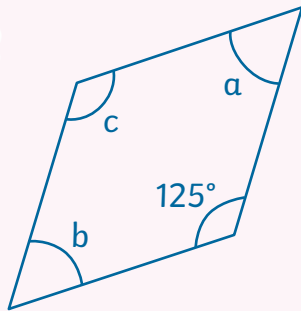
Not to scale

Angles in Quadrilaterals

Use your knowledge of angles to find the missing angles without using a protractor.

The interior angles of a quadrilateral always add up to 360° .

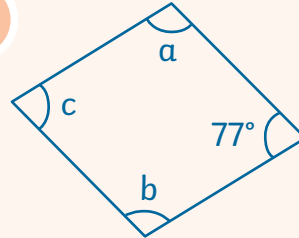
1



$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

$$c = \underline{\hspace{2cm}}$$

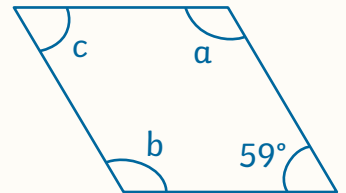
2



$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

$$c = \underline{\hspace{2cm}}$$

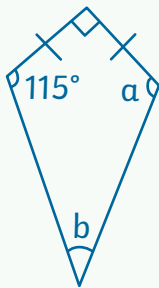
3



$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

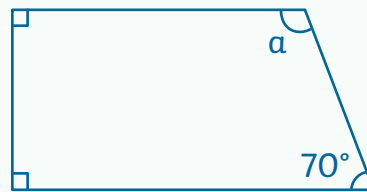
$$c = \underline{\hspace{2cm}}$$

4



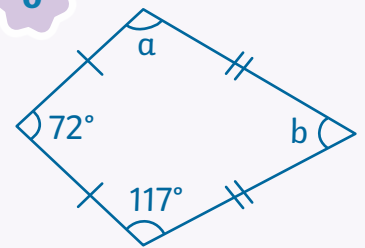
$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

5



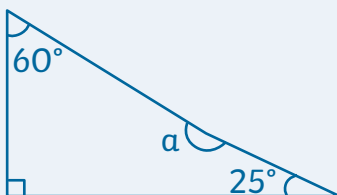
$$a = \underline{\hspace{2cm}}$$

6



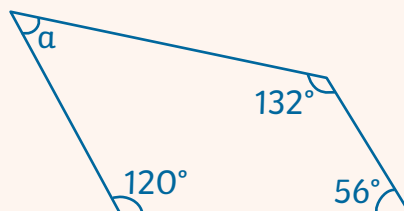
$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

7



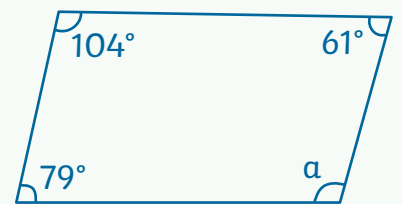
$$a = \underline{\hspace{2cm}}$$

8



$$a = \underline{\hspace{2cm}}$$

9



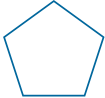
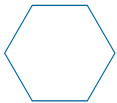
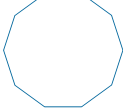
$$a = \underline{\hspace{2cm}}$$

Not to scale

Angles in Regular Polygons

Use your knowledge of angles to complete the table, then find the missing angles without using a protractor.

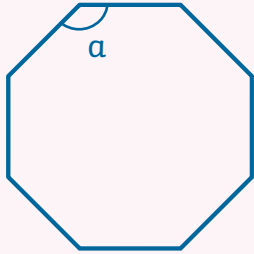
The sum of the interior angles of a regular polygon can be found by solving $(n - 2) \times 180^\circ$.

Shape	Number of Sides (n)	Formula $(n - 2) \times 180^\circ$	Sum of Interior Angles
square 	4	$(4 - 2) \times 180^\circ$	360°
regular pentagon 			
regular hexagon 			
regular heptagon 			
regular octagon 			
regular nonagon 			
regular decagon 			

Angles in Regular Polygons

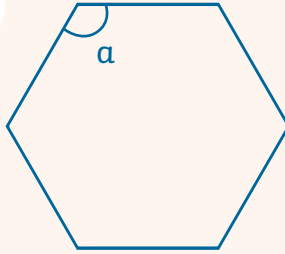
Once we know the sum of the interior angles of a regular polygon, to find the size of a single angle we can divide the sum by the number of sides.

1



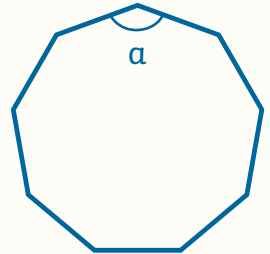
$a =$ _____

2



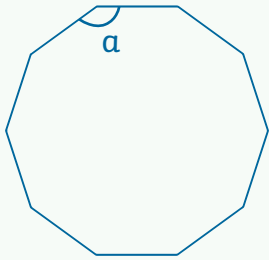
$a =$ _____

3



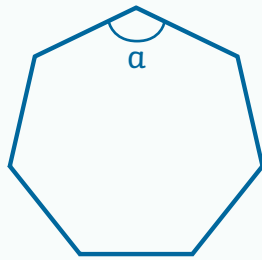
$a =$ _____

4



$a =$ _____

5



$a =$ _____

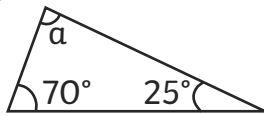
Challenge:

On a regular 24-sided shape, what would be the size of each interior angle?

Not to scale

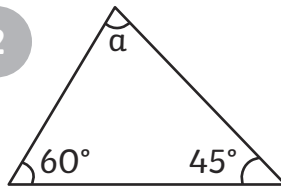
Angles in Triangles Answers

1



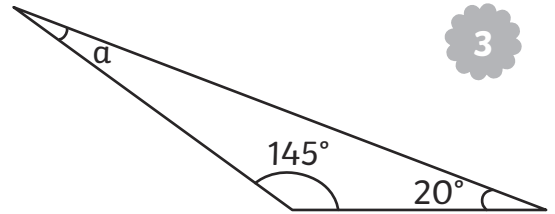
$$a = \underline{85^\circ}$$

2



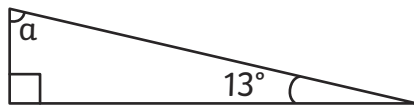
$$a = \underline{75^\circ}$$

3



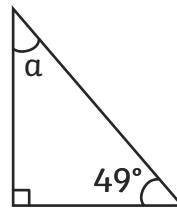
$$a = \underline{15^\circ}$$

4



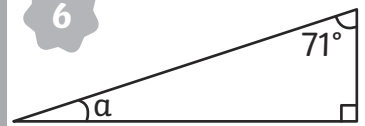
$$a = \underline{77^\circ}$$

5



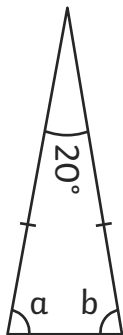
$$a = \underline{41^\circ}$$

6



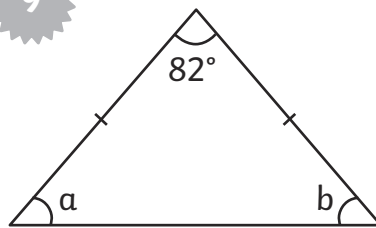
$$a = \underline{19^\circ}$$

8



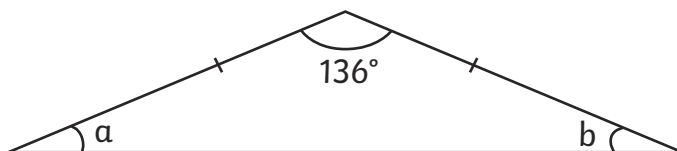
$$a = \underline{80^\circ} \quad b = \underline{80^\circ}$$

9



$$a = \underline{49^\circ} \quad b = \underline{49^\circ}$$

10

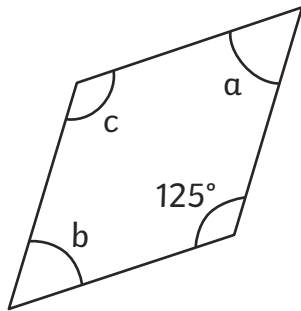


$$a = \underline{22^\circ} \quad b = \underline{22^\circ}$$

Not to scale

Angles in Quadrilaterals Answers

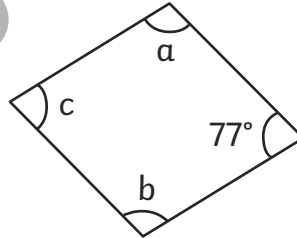
1



$$a = \underline{55^\circ} \quad b = \underline{55^\circ}$$

$$c = \underline{125^\circ}$$

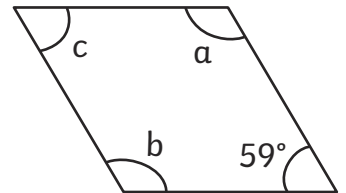
2



$$a = \underline{103^\circ} \quad b = \underline{103^\circ}$$

$$c = \underline{77^\circ}$$

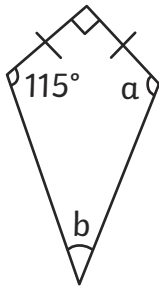
3



$$a = \underline{121^\circ} \quad b = \underline{121^\circ}$$

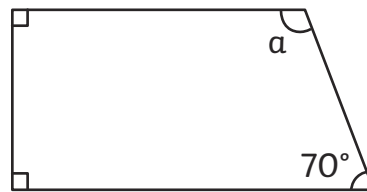
$$c = \underline{59^\circ}$$

4



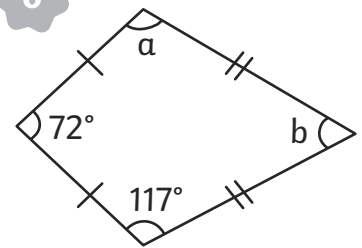
$$a = \underline{115^\circ} \quad b = \underline{40^\circ}$$

5



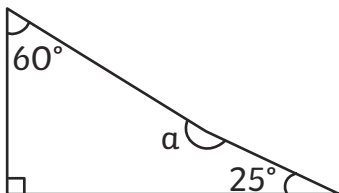
$$a = \underline{110^\circ}$$

6



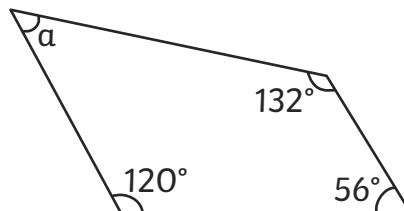
$$a = \underline{117^\circ} \quad b = \underline{54^\circ}$$

7



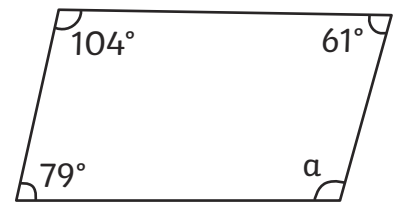
$$a = \underline{185^\circ}$$

8



$$a = \underline{52^\circ}$$

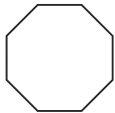
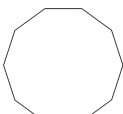
9



$$a = \underline{116^\circ}$$

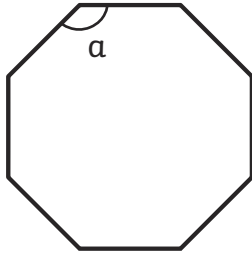
Not to scale

Angles in Regular Polygons Answers

Shape	Number of Sides (n)	Formula (n - 2) x 180°	Sum of Interior Angles
square 	4	$(4 - 2) \times 180^\circ$	360°
regular pentagon 	5	$(5 - 2) \times 180^\circ$	540°
regular hexagon 	6	$(6 - 2) \times 180^\circ$	720°
regular heptagon 	7	$(7 - 2) \times 180^\circ$	900°
regular octagon 	8	$(8 - 2) \times 180^\circ$	1080°
regular nonagon 	9	$(9 - 2) \times 180^\circ$	1260°
regular decagon 	10	$(10 - 2) \times 180^\circ$	1440°

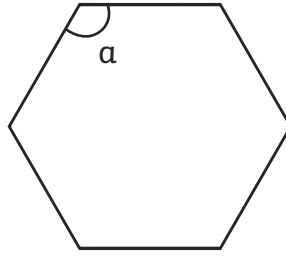
Angles in Regular Polygons Answers

1



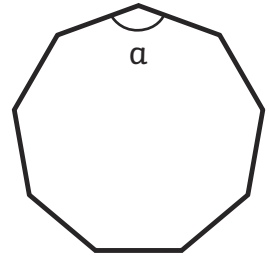
$$\alpha = \underline{135^\circ}$$

2



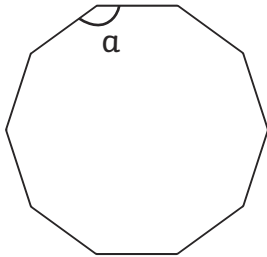
$$\alpha = \underline{120^\circ}$$

3



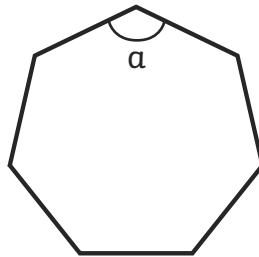
$$\alpha = \underline{140^\circ}$$

4



$$\alpha = \underline{144^\circ}$$

5



$$\alpha = \underline{128.6^\circ}$$

Challenge:

On a regular 24-sided shape, what would be the size of each interior angle?

165°

Not to scale

We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.